Data Lake Solution Architecture Diagram

On-Premise Hadoop Ecosystem · HIPAA Compliant · Horizontally Scalable

CLIENTS

Medical Facility
8,000 sites

FTP / SFTP
File Transfers

REST APIs
External Systems

Legacy SQL
SQL Server Migration

File Formats
CSV · XML · TXT
15M files/day

Data Volume
77,000 zip files/day · 3,500/hr
15–20% YoY growth

End Users
Analysts · Engineers · Customers
BI · Ad-hoc · Reporting

INGESTION

① INGESTION LAYER

Apache NiFi
Visual Dataflow · FTP Polling
Gzip Decompression · Validation

Apache Kafka
Message Queue · 3,500 files/hr
Topic Partitioning · Replay

Debezium
Change Data Capture (CDC)
SQL Server Logs · UPSERT

Apache Sqoop
Bulk DB Import · SQL → HDFS
Parallel Import · Incremental

Apache Flume
Log & File Collection
Agent-Based · Reliable Delivery

Apache Airflow
Orchestration · Scheduling
Pipeline Monitoring

STORAGE

② STORAGE LAYER — HDFS (Replication Factor: 3)

Raw Zone
Landing Area · Immutable
Parquet / ORC · Audit Trail

Staging Zone
Cleaned & Validated
Apache Hudi · UPSERT · CDC

Curated Zone
Gold · Analytics-Ready
Parquet · Partitioned · 5yr+

Archive Zone
Cold Storage · HIPAA 7yr+
SSD → HDD → JBOD Tiering

Metadata Store
Apache Atlas · Data Catalog
Lineage · PII/PHI · Hive Metastore

PROCESSING

③ PROCESSING LAYER

Batch Processing
Apache Spark (PySpark) · YARN
Metadata-Driven ETL · Replaces SSIS

Stream Processing
Kafka Streams · Apache Flink
Near-Real-Time · Sub-minute Latency

Ad-hoc & CDC
Hive · Impala · Presto / Trino
Federated SQL · Time-Travel Queries

ML & Analytics
Spark MLlib · TensorFlow · MLflow
Risk Scoring · Experiment Tracking

SERVING

④ SERVING LAYER

SQL Reports
Hive · Impala
Daily · Weekly · Compliance

Dashboards
Apache Superset · Grafana
PowerBI · Tableau · Real-time

Web Portal & APIs
FastAPI · Customer Self-Service
Bed Availability · Load Balanced

ML Model Serving
MLflow REST · Predictions
Risk Scores · New Capability

Ad-hoc Analytics
Presto / Trino · Jupyter Notebooks
Federated SQL · Eliminates Silos

CROSS-CUTTING

⑤ CROSS-CUTTING CONCERNS — Spans All Layers

Security & Governance
Apache Ranger (RBAC) · Knox Gateway
Kerberos Auth · AES-256 · TLS/SSL
HIPAA Compliance · PHI Classification

HA & Disaster Recovery
HDFS Replication Factor: 3
NameNode HA · ZooKeeper Failover
Apache DistCp · Sub-minute RTO

Monitoring & Operations
Ambari / Cloudera Manager
Prometheus · Grafana · ELK Stack
Airflow UI · Pipeline Alerts

Hadoop Cluster
Cloudera HDP · On-Premise
YARN Resource Management
Horizontal Scale-out · No Vendor Lock

Ingestion · Storage · Processing · Serving · Cross-Cutting · Data Flow

Stack: NiFi · Kafka · Debezium · Sqoop · Flume · HDFS · Hudi · Atlas · Spark · Flink · Hive · Impala · Presto · Airflow · Ranger · Knox · ZooKeeper · Superset